SREENARAYANAGURU OPEN UNIVERSITY

The State University for Education, Training and Research in Blended Format, Kerala

ASSIGNMENT SUBMISSION FORM - I

(To be submitted by Learners along with the handwritten assignment)

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Name of the Programme	: BCA
Semester	: ~~THIR~~ FOURTH
Course Code	: B21CA08DC
Course Title	: SOFTWARE ENGINEERING
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Signature of the Candidate with date

(To be filled in by the Evaluators)

Sl.No	Evaluation Criteria	Maximum Marks	Marks Awarded
1	Introduction	2	
2	Explanation of the theory or idea	6	
3	Overall clarity, style, structure and organization	3	
4	Clear Conclusion	2	
5	Bibliography or Reference	2	
	Total	15	

Name of the Academic Counsellor:

Signature:

Date:

ASSIGNMENT

Assignment Question: Describe the concept of Devops. explaining its definition, need and key benefit in modern software development Also, outline the 7 c's of Devops and list commonly used Devops tools

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INTRODUCTION

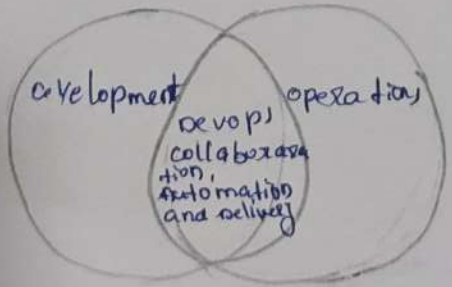
In today's fast-paced digital world, businesses need to release software updates quickly and reliably. In the past, the software process was slowed down by a big divide between the developers who wrote the code and the operations teams who managed the servers. Developers wanted to push new features fast, while operations wanted to keep systems stable, which often led to miscommunication, bugs, and delayed releases.

To fix this problem, modern tech companies use a method called DevOps, which combines software development and IT operations. DevOps breaks down these team walls by focusing on automation, shared goals, and continuous teamwork. By using the right automated tools throughout the software lifecycle, DevOps allows companies to test and deploy software faster, catch errors early, and deliver a better experience to users.

- What is Devops

The process of defining what is devops often requires looking at it as a culture method, and tools for improving the flow of software development. one of the key ideas of Devops is to combine the roles of the developers who write code with operations team who deploy and maintain it to release code faster and receive feedback more frequently. As a result, Devops introduces an organizational rhythm by focusing on automation, iterative work, and shared responsibility.

The term Devops is not only a set of tools, it is a culture of collaboration between people, processes and technology, in processes ranging from server setup to the implementation of small code changes into production.



Devops breaks down the silos. Developers and operations work together from the start to end of the lifecycle of an application. They collaborate on everything from code creation to deployment and monitoring. for example, developers write the code and they collaborate with operations to set up the correct infrastructure to deploy the application effectively

For example, if developer write code for a new feature, they might work closely with the operations team to ensure that the infrastructure is properly set up for smooth deployment. They may even use shared tools like Docker, to ensure consistency across environments (development, testing, and production)

- Need of DevOps

In a traditional software development environment, developers and IT operations team often work seperately. Developers write code, while operations teams are responsible for deploying and maintaining the infrastructure. This division can lead to communication gaps, delays. an This creates a "wall" between the teams because developers want speed while operations want stability. As a result, there's often a lot of miscommunication. and things can slow down.

~~How fix~~

How DevOps fixes This :-

DevOps helps by bringing developers and operations together to work as one team the goal is to deliver high-quality software quickly while keeping things stable. see how it fixes the problem.

Working Together :- Developers and operations teams collaborate closely from the start, so both team understand each other's goals.

Shared responsibility :- Both team are responsible for making sure the software works well and gets to user quickly

Automation :- Devops uses automated tools speed up the process of building testing, and deploying software, so things are faster and more reliable.

Devops also solves some common problems in traditional software development like:

1. Slow Deployments.

Deploying new features or fixes can take a long time because it's often a manual process. The solutions are, Automation speed up the process code is automatically built, tested, and deployed without much human intervention, making releases faster.

2. Manual Errors

People sometimes make mistake when doing things by hand, like configuring servers or deploying code.

Devop · Devops uses automated tools that handle tasks without human error, making things more consistent.

3. Downtime during updates.

when updates are made, they can cause the application to go offline, disrupting users. with Devops, updates can be done in a way that avoids downtime For example- they use smart techniques to updated parts of the system without affecting the entire app.

4. Poor Communication

Devop developers and operations teams often don't communicate well, which can lead to misunderstandings and delays. Collaboration is key. The teams communicate regularly and use shared tools to stay on the same page. ensuring everyone understands what need to be done.

- Key Benefits of Devops

→ Quantifiable Gains :-

- Improve collaboration :-
  It promotes better communication and

cooxpxation between development and opexations teams, leading to inexeased pxoductivity and effiuency

- Enhanced soft wxxe quality.

continuous testing and monitoxing help identify and xesolve defects early, xesulting in moxe Reliable and high-quality applications.

- Highex Deploy ment Success Rate:

This is the top moxit of oevops that deploy-ment becomes moxe xeliable and stable. failuxe in deployment in txadional way is often attxibuted to different causes like envixonment mismatch. missing dependencies, ox last, minute configuxation issues.

- Reduction in Development Time.

oevops significantly xeduces development time by automating pxocesses such as code integxation, testing and deployment. continuous integxation and continuous delivexy pxactices enable development teams to detect and fix issues early, stxeamline wxxkflows

and release software update more frequently. As a result, organizations can accelerate the software development lifecycle, delivery products to market faster, and respond more quickly to changing customer requirements.

Business impact:

• Faster Time to Market

Devops enables organizations to deliver application features, and updates to customers more quickly by automating development, testing, and deployment processes. Through continuous integration and continuous delivery, teams can release software more frequently and efficiently. This faster time to market helps businesses respond rapidly to changing customer needs, gain a competitive advantage and increases customer satisfaction.

• Reduced Operational Costs.

Devops reduces operational costs by automating tasks and improving efficiency, resulting in lower maintenance and infrastructure expenses

• Higher Customer satisfaction

By delivering high-quality software faster and responding rapidly to user needs, DevOps helps organization provide a better customer experience.

- 7 C's of Devops

They believe that with the right people, it's possible to build the right processes which then lead to choosing the right tools to achieve the desired results. The 7 c's of Devops are:-

1. communication

Good Devops starts with clear and open communication between developers, testers, operations and other teams. Everyone should know what's happening (what's expected, and what needs fixing

2. Collaboration

All teams work together instead of doing their jobs seperately. Developers, testers and ops must share responsibilities and support each other. Example: A developer checks with the testing team

before adding a new feature to ensure it won't break anything.

3. controlled process.

There should be a clear and consistent way of doing things like how to develop, test, and release software. This avoid confusion and mistakes. eg:- every code change follows a set process: code - review - test - deploy.

4. continuous integration.

Developers keep adding small bits of code regularly and the system automatically check if everything still works, This helps catch errors early eg:- developer pushs code, automatic test run to make sure nothing is broken.

5. Continuous Deployment (CD)

once the code is tested, it is automatically released to production without delay This helps in fast and safe updates. eg:- A new feature is automatically made live after passing all tests - no need.

6. continuous Testing.

Testing happen all the time using automated tools This ensures the software is always in good shape. eg:- Automated tests check the login page still work every time Someone changing code.

7. Continuous Monitoring.

The system is watched all the time to make sure it runs smoothly. Alerts are sent if something goes wrong. Example: A monitoring tool tracks how much memory a server is using and warns the team before it crashes

- Devops Tools

Automation is the heart of Devops. Here some tools used in various stages

Stage	Tools.
Provisioning	Terraform, Ansible, chef, puppet, Aws cloud formation
CI	Jenkins, Gitlab ci, Circle ci, Travis ci, Bamboo.
CD	Spinnaker, ArgoCD, GitHub Actions, flux, octopus Deploy.
Testing	selenium, Junit, TestNG, pyTest, postman, cucumber
Monitoring	prometheus, Grafana Nagios, Datadog New Relic, ELK Stack (Elastic Search, Logstash, Kibana

CONCLUSION

Devops has become a vital approach in modern Software development, helping organizations improve collaboration between development and operations teams while increasing automation and efficiency By enabling faster delivery of high quality, software and enhancing system reliability. Devops allows businesses to respond effectively to changing customer demands and remain competitive in today's fast-paced digital environment.

In addition, the 7c's of Devops and various DevOps tools provide a structured framework for continuous development, testing, deployment, and monitoring. These practices help reduce development time and operational costs while improving overall software quality and customer satisfaction. As a result, DevOps plays a key role in the successful development and maintenance of modern software applications.

REFERENCE

- Https://www.spxintzeal.com/blog/what-is-devops
- Https://aws.amazon.com/devops/what-is-devops/
- https://www.sentinelone.com/cybasecurity-101/cybersecurity/what-is-devops/
- https://jfrog.com/learn/devops/devops/
- https://azure.microsoft.com/en-in/resources/cloud computing-dictionary/what-is-devops